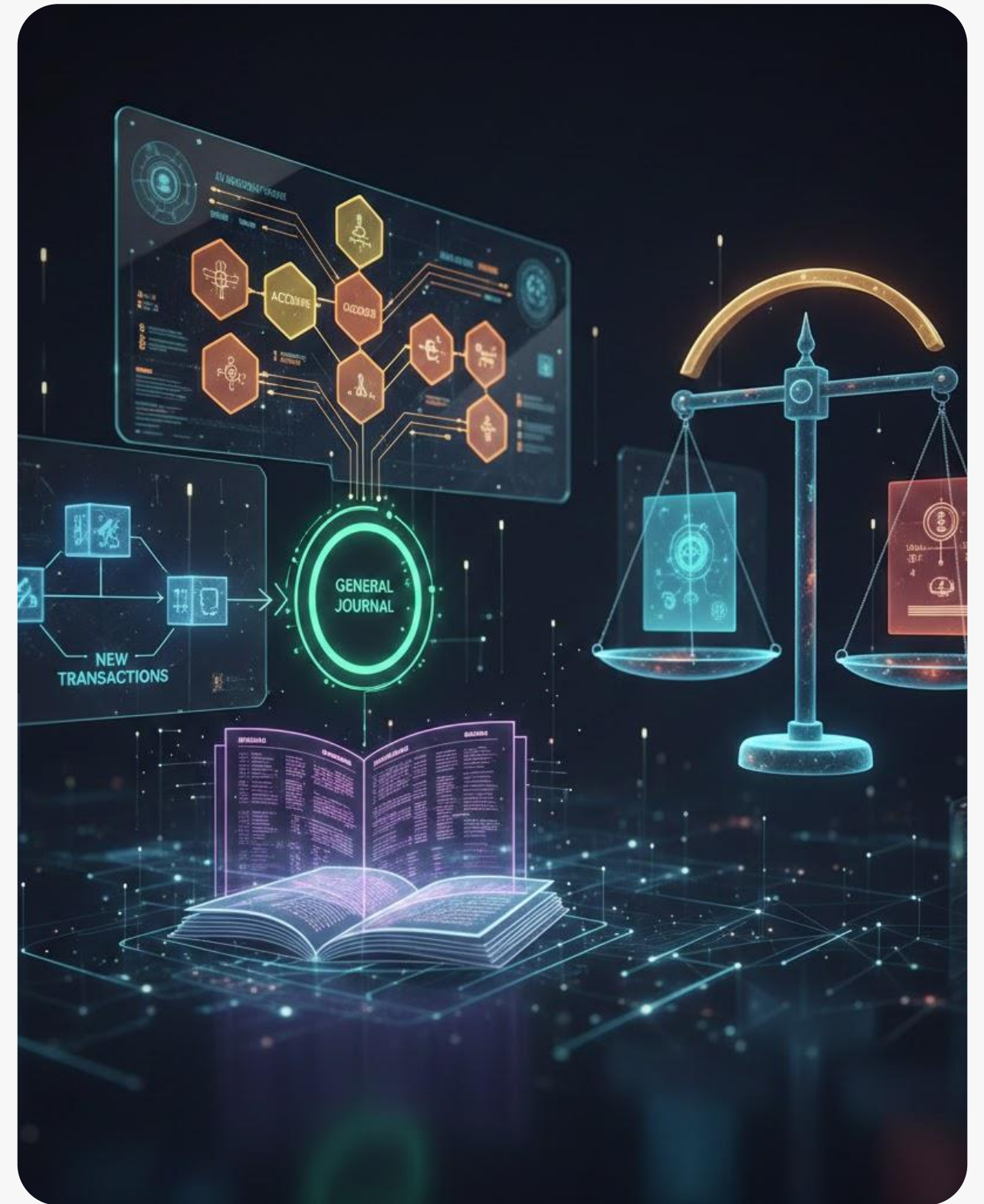
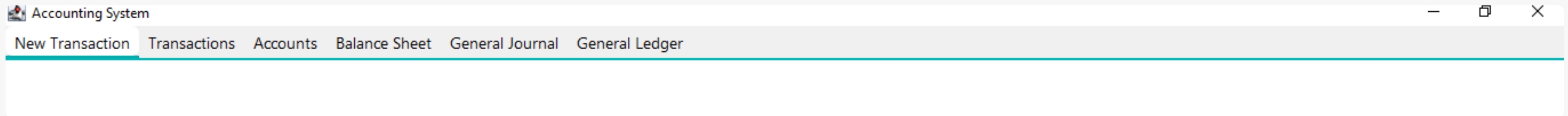


ACCOUNTING SYSTEM

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System Overview

A Java Swing-Based Accounting Management System

- Records transactions and updates all accountingpanels automatically
- Covers complete accounting cycle

Main panels of our system:

1. New Transaction
- 2.Accounts
- 3.General Journal
- 4.General Ledger
- 5.Balance Sheet

These panels are connected.

When we add a transaction, all panels update automatically.

```
public AccountingSystem() {  
    setTitle("Accounting System");  
    setSize(1500, 800);  
    setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);  
    setLocationRelativeTo(null);  
    setLayout(new BorderLayout());  
}
```

```
mainPanel = new JPanel(new CardLayout());  
mainPanel.add(newTransactionPanel, "New Transaction");  
mainPanel.add(transactionsPanel, "Transactions");  
mainPanel.add(accountsPanel, "Accounts");  
mainPanel.add(balanceSheetPanel, "Balance Sheet");  
mainPanel.add(generalJournalPanel, "General Journal");  
mainPanel.add(generalLedgerPanel, "General Ledger");
```

New Transaction Panel

- User inputs Date, Description, Debit, Credit, Amount
- Dropdowns for selecting accounts
- Saves transaction and updates all panels
- The design is simple and clean,

Below is the code snippet of the design

```
// New Transaction Panel
private JPanel createNewTransactionPanel(Accounts accountsPanel,
    BalanceSheet balanceSheetPanel,
    GeneralJournal generalJournalPanel,
    GeneralLedger generalLedgerPanel) {

    JPanel panel = new JPanel(new BorderLayout());
    panel.setBackground(Color.WHITE);

    JPanel formPanel = new JPanel(null);
    formPanel.setPreferredSize(new Dimension(700, 350));
    formPanel.setBackground(Color.WHITE);

    JPanel topCenterWrapper = new JPanel(new FlowLayout(FlowLayout.CENTER, 0, 40));
    topCenterWrapper.setBackground(Color.WHITE);
    topCenterWrapper.add(formPanel);
    panel.add(topCenterWrapper, BorderLayout.NORTH);

    JLabel titleLabel = new JLabel("Add a New Transaction");
    titleLabel.setFont(new Font("Segoe UI", Font.BOLD, 20));
    titleLabel.setForeground(new Color(0, 150, 150));
    titleLabel.setBounds(107, 13, 250, 30);
    formPanel.add(titleLabel);
```

Add a New Transaction

	Date :	2025-12-09
	Description :	
	Debit Account :	Cash [ASSET] ▼
	Credit Account :	Cash [ASSET] ▼
	Amount :	
		Add Transaction Clear

New Transaction Code Snippet

- Handles the process of recording new transactions into the system.
- Ensures every entry is properly validated, preventing wrong or incomplete data.
- Validation for date, amount, and accounts
- This prevents the app from crashing when the user inputs invalid values

```
// Validate date
try {
    LocalDate.parse(date);
} catch (DateTimeParseException ex) {
    showStatus(statusLabel, "X Invalid date format (YYYY-MM-DD)", Color.RED);
    return;
}

// Validate amount
try {
    amount = Double.parseDouble(amountField.getText().trim());
    if (amount <= 0) {
        showStatus(statusLabel, "X Amount must be greater than 0", Color.RED);
        return;
    }
} catch (NumberFormatException ex) {
    showStatus(statusLabel, "X Invalid input. Please check all fields", Color.RED);
    return;
}

// Validate accounts
if (debit.equals(credit)) {
    showStatus(statusLabel, "X Debit and Credit accounts cannot be the same", Color.RED);
    return;
}

Transaction t = new Transaction(date, desc, debit, credit, amount);
transactions.add(t);
refreshTransactionTable("");
```

Accounts Panel

- Shows list of all accounts
- Includes: Assets, Liabilities, Equity, Revenue, Expenses
- Simple table layout

Accounting System

New Transaction Transactions **Accounts** Balance Sheet General Journal General Ledger

Protect Against Inflation

Accounts

List of all ledger accounts and current balances

Account	Type	Balance
Cash	ASSET	0.00
Accounts Receivable	ASSET	0.00
Inventory	ASSET	0.00
Prepaid Expenses	ASSET	0.00
Equipment	ASSET	0.00
Accumulated Depreciation	ASSET	0.00
Accounts Payable	LIABILITY	0.00
Notes Payable	LIABILITY	0.00
Owner's Capital	EQUITY	0.00
Service Revenue	INCOME	0.00
Sales Revenue	INCOME	0.00
Rent Expense	EXPENSE	0.00
Salaries Expense	EXPENSE	0.00
Supplies Expense	EXPENSE	0.00
Utilities Expense	EXPENSE	0.00
Cost of Goods Sold	EXPENSE	0.00

Achieve Life Goals

Accounts Code Snippet

- This panel shows all accounts with their type and current balance.
- Users can also add new accounts.
- This creates the table for displaying account names, types, and balances.

```
String[] columns = {"Account", "Type", "Balance"};
model = new DefaultTableModel(columns, 0) {
    private static final long serialVersionUID = 1L;
    @Override public boolean isCellEditable(int row, int column) { return false; }
};

table = new JTable(model);
table.setShowGrid(false);
table.setInterCellSpacing(new Dimension(0, 0));
table.setBackground(Color.WHITE);
table.setForeground(Color.BLACK);
table.setFont(new Font("Segoe UI", Font.PLAIN, 13));
table.setRowHeight(25);
table.setFillViewportHeight(true);
```


General Journal Panel

[illegible]

- Records all transactions
- Debit + Credit in journal format
- Auto-formatted rows

General Ledger Panel

- All accounts have their own ledger
- Debit and credit posting is automatic
- Running balance displayed
- Simple Scrollbar Design

Account: Cash

Date	Description	Debit	Credit	Balance
	Cash			
	Accounts Receivable			
	Inventory			
	Prepaid Expenses			
	Equipment			
	Accumulated Depreciation			
	Accounts Payable			
	Notes Payable			

[illegible]

Balance Sheet

- Organized into Assets, Liabilities, Equity
- Auto calculates totals
- Ensures Assets = Liabilities + Equity

Accounting System		— □ ×	
New Transaction		Transactions	
Accounts		Balance Sheet	
General Journal		General Ledger	
Balance Sheet		Print	
Statement of assets, liabilities, and equity			
Assets		Liabilities & Equity	
Account	Amount	Account	Amount
Cash	0.00	Accounts Payable	0.00
Accounts Receivable	0.00	Notes Payable	0.00
Inventory	0.00	Owner's Capital	0.00
Prepaid Expenses	0.00		
Equipment	0.00		
Accumulated Depreciation	0.00		
Supplies Expense	0.00		
Total Assets	0.00	Total Liabilities + Equity	0.00

Features & Functionalities

- ✓ Auto calculation
- ✓ Real-time ledger posting
- ✓ Clean UI and organized layout
- ✓ Input validation
- ✓ Accounting-standard formats
- ✓ Accurate running balance display
- ✓ Auto totals for all statements

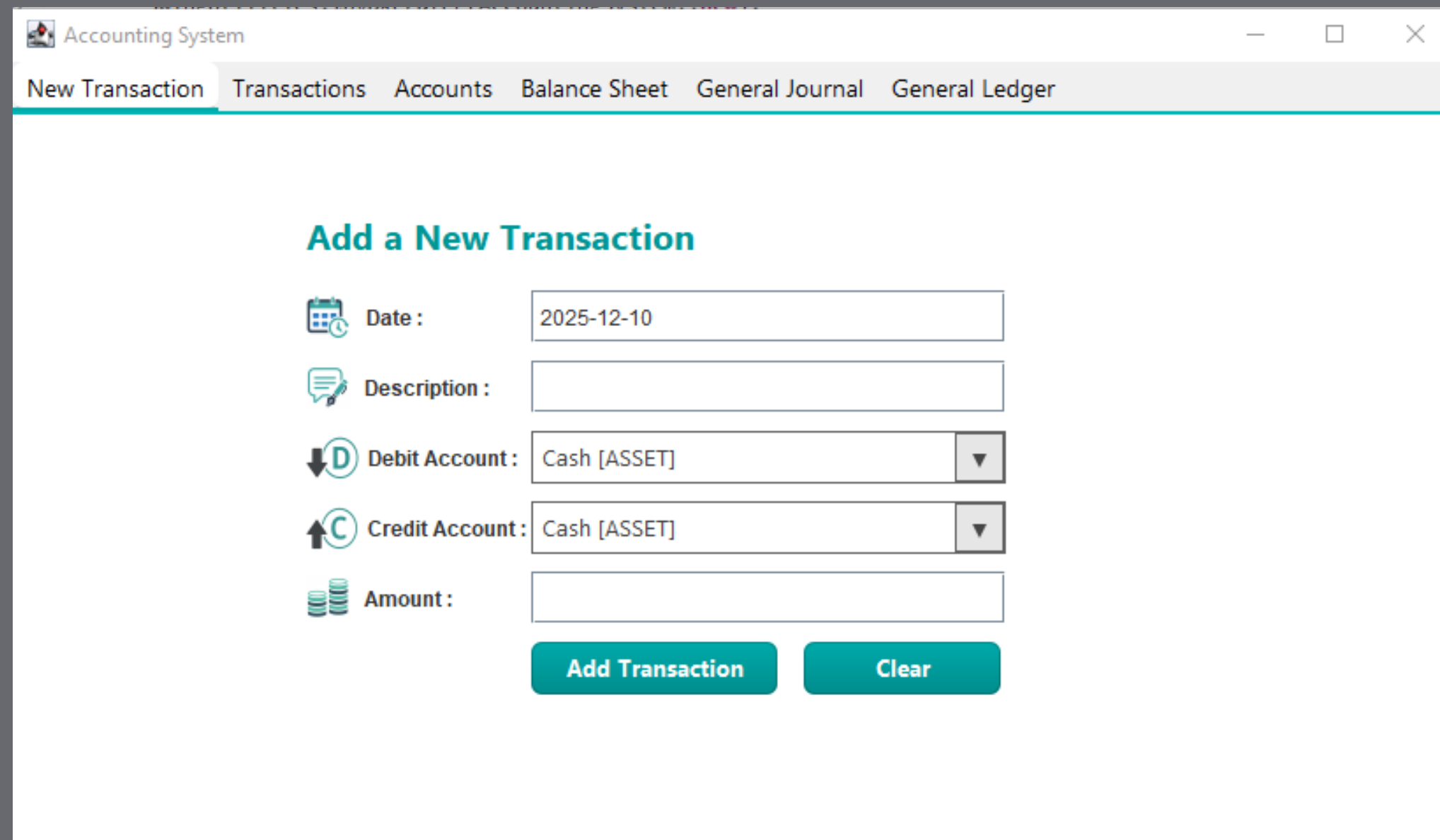
This is the central method that updates the Accounts, Journal, Ledger, and Balance Sheet.

```
// Create feature panels
Accounts accountsPanel = new Accounts();
BalanceSheet balanceSheetPanel = new BalanceSheet(accountsPanel);
GeneralJournal generalJournalPanel = new GeneralJournal();
GeneralLedger generalLedgerPanel = new GeneralLedger();

// Link cross-panel updates
accountsPanel.setBalanceSheetListener(balanceSheetPanel);
accountsPanel.setGeneralJournalListener(generalJournalPanel);
accountsPanel.setGeneralLedgerListener(generalLedgerPanel);
```

- The system connects all panels together.
- When a transaction is added, all modules update automatically.
- This prevents errors and ensures accuracy.

Conclusion



The screenshot displays a web application titled "Accounting System" with a navigation bar containing "New Transaction", "Transactions", "Accounts", "Balance Sheet", "General Journal", and "General Ledger". The "New Transaction" tab is active. Below the navigation bar, the form "Add a New Transaction" is visible. It includes the following fields and controls:

- Date :** A text input field containing "2025-12-10".
- Description :** A text input field.
- Debit Account :** A dropdown menu with "Cash [ASSET]" selected.
- Credit Account :** A dropdown menu with "Cash [ASSET]" selected.
- Amount :** A text input field.
- Buttons:** "Add Transaction" and "Clear".

- Accounting system is simple, functional, and accurate.
- It follows accounting rules and Java OOP structure.
- It automates manual accounting processes.